

Program: Diploma in Fire Technology and Safety	
Course Code: 3469	Course Title: Electrical Safety Engineering Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To understand the importance of electrical safety in various electrical systems.
- To learn the principles of protection and safety mechanisms in electrical circuits.
- To familiarize with safety standards and testing procedures for electrical installations.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of electrical quantities and their units		Fundamentals of Electrical and Electronics Engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Identify and analyze different electrical safety devices and systems to ensure safety in electrical installations.	11	Applying
CO 2	Understand and implement overload protection, short-circuit protection, and other safety measures in electrical circuits.	10	Applying
CO 3	Examine the safety mechanisms required for high-voltage systems and their protection under fault conditions.	10	Applying
CO 4	Perform practical tests and simulations to verify electrical safety protocols and mechanisms in various electrical systems.	10	Applying
	Lab Exam	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	2	1	2	3
CO2	3	3	2	2	2	2	2
CO3	2	2	3	3	2	2	1
CO4	2	3	2	2	3	2	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify and analyze electrical safety devices and systems		
M1.01	Investigate overload protection in transformers under different load conditions..	2	Applying
M1.02	Analyze short-circuit protection devices in electrical circuits.	3	Applying
M1.03	Simulate electrical hazards in wet environments and assess leakage currents.	3	Applying
M1.04	Test earth leakage circuit breakers (ELCBs) and evaluate their performance.	3	Applying
CO2	Understand and implement various electrical protection mechanisms		
M2.01	Evaluate fuse and circuit breaker coordination for safety during faults.	2	Understanding
M2.02	Analyze high-voltage testing procedures for cable insulation.	3	Understanding
M2.03	Investigate the effectiveness of lightning protection systems during electrical storms.	3	Understanding
M2.04	Simulate fault conditions in a power grid and assess the protection mechanisms.	2	Applying
	Lab Exam I	2	
CO3	Examine the safety protocols required for high-voltage systems		
M3.01	Understand high-voltage insulation testing for electrical cables..	3	Understanding

M3.02	Analyze the operation and coordination of overcurrent protection devices under fault conditions.	2	Applying
M3.03	Investigate and evaluate lightning protection systems in electrical systems.	2	Applying
M3.04	Simulate power grid faults and verify the safety protocols during failure conditions.	3	Applying
CO4	Verify and test electrical safety systems and mechanisms		
M4.01	Test overload and short-circuit protection devices using standard testing methods.	2	Applying
M4.02	Analyze the effectiveness of earthing systems in different electrical setups.	2	Understanding
M4.03	Evaluate the response of protection devices like fuses and circuit breakers under fault conditions.	3	Applying
M4.04	Assess the function of earth leakage devices in preventing electrical hazards.	3	Understanding
	Lab Exam – II	2	

Text / Reference:

T/R	Book Title/Author
T1	Electrical Safety Engineering - J. C. Das (McGraw-Hill Education).
T2	Electrical Safety Handbook - H. P. S. Kalsi (McGraw-Hill Education).
R1	Electrical Safety: A Guide to Safe Installation and Operation - Dennis K. McBride.
R2	Fundamentals of Electrical Safety - A. K. Gupta (Khanna Publishers).